Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	4050	**3**	**M1** for $\frac{\text{figs } 89}{\text{figs } 22}$ **A1** for 4045[. …] or $4045\frac{5}{11}$ **B1** for *their* answer seen (but not a multiple of 10) correctly rounded to the nearest ten, provided does not round to zero
1(a)(ii)	132	**2**	**M1** for 4650 ÷ 35
1(b)	676[.00]	**1**	
1(c)	227.5[0] 97.5[0] 65[.00]	**3**	**M2** for 390 ÷ (7 + 3 + 2) × k or better where k is 7, 3 or 2 or **M1** for 390 ÷ (7 + 3 + 2) or better, implied by 32.5[0]
1(d)	12	**3**	**M2** for $\frac{3500-3080}{3500}$ [× 100] or $(1-\frac{3080}{3500})$ [× 100] or [100 –] $\frac{3080}{3500}$ × 100 or **M1** for $\frac{3080}{3500}$ or 3500 – 3080
1(e)	Fully correct net	**2**	**B1** for at least two correct faces drawn in the correct place
2(a)(i)	Two correct bars	**1**	
2(a)(ii)	3	**1**	
2(b)	68 [h] 15 [min]	**3**	**B1** for 9 [h] 45 [m] or $9\frac{3}{4}$ or 9.75 or 585 seen **M1** for *their* (9 [h] 45 [m], $9\frac{3}{4}$, 9.75 or 585) × 7 soi

Question	Answer	Marks	Partial Marks
2(c)	424	**3**	**M2** for 24 × 10.5 + 16 × 6.25 + 8 × 9 soi or **M1** for two of 24 × 10.5, 16 × 6.25, 8 × 9
2(d)	30	**1**	
2(e)(i)	21 05 or 9.05 pm	**1**	
2(e)(ii)	20 20 or 8.20 pm	**2**	**M1** for 23 15 [− (10 min)] − (2 h 45 m) soi by 20 30
2(f)	1.45	**2**	**M1** for [5 −] (1.85 + 1.70) or 3.55
2(g)(i)	21	**1**	
2(g)(ii)	–14	**1**	
3(a)(i)	$12c - d$ final answer	**2**	**B1** for $12c$ or $-d$
3(a)(ii)	$8x^{10}$ final answer	**2**	**B1** for $8x^n$ or kx^{10} (n and $k \neq 0$)
3(b)	-1.5 or $-1\frac{1}{2}$ oe nfww	**2**	**M1** for $-2 - 1 = 5x - 3x$ or better or $3x - 5x = 1 + 2$ or better
3(c)	$xy(3x - 5)$ final answer	**2**	**B1** for $y(3x^2 - 5x)$ or $x(3xy - 5y)$ or correct answer spoilt
3(d)	$[r =]\ \frac{T}{3} - 5$ oe nfww final answer	**2**	**M1** for first correct step e.g. $T = 3r + 15$ or $\frac{T}{3} = r + 5$
4(a)	Enlargement [centre] (6, –5) [sf] 2	**3**	**B1** for each
4(b)	Rotation [centre] (0, 0) oe 180°	**3**	**B1** for each
4(c)	Correct translation to (3, 3), (3, 0), (5, 0)	**2**	**B1** for a translation by $\begin{pmatrix} 2 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 5 \end{pmatrix}$
4(d)	Correct reflection to (3, –5), (5, –2),(5, –5)	**2**	**B1** for correct reflection in $x = k$
5(a)	Two correct lines	**2**	**B1** for 1 correct line and no diagonals or 2 correct lines and one diagonal
5(b)	120	**3**	**M2** for 180 − (360 ÷ 6) oe or (6 – 2) × 180 ÷ 6 oe or **M1** for 360 ÷ 6 oe or (6 – 2) × 180 oe